

Clustering in Unsupervised Learning

K-Means • Hierarchical • DBSCAN • Gaussian Mixture Model

Machine Learning YouTube Series

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1. Introduction to Clustering

Clustering is a fundamental technique in **Unsupervised Learning** where similar data points are grouped together into clusters. Unlike supervised learning, clustering does not use labeled data — the algorithm discovers hidden structures and patterns on its own.

Applications of Clustering

- Customer Segmentation
- Recommendation Systems
- Image Segmentation
- Fraud Detection
- Social Network Analysis
- Medical Diagnosis

What Is Clustering?

Clustering groups similar data points together while separating dissimilar points into different groups. For example, customers with similar shopping behaviour belong to the same cluster; news articles with similar topics can be grouped together.

2. Types of Clustering Algorithms

The four main clustering algorithms covered in this document:

1. K-Means Clustering
2. Hierarchical Clustering
3. DBSCAN (Density-Based)
4. Gaussian Mixture Model (GMM)

3. K-Means Clustering

K-Means is one of the most popular clustering algorithms. It divides a dataset into **K clusters** where each data point belongs to the cluster with the nearest centroid.

How K-Means Works

Step 1 — Choose K	Select the number of clusters K.
Step 2 — Initialize Centroids	Randomly place K centroids in the feature space.
Step 3 — Assign Data Points	Assign each point to the nearest centroid using Euclidean distance: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Step 4 — Update Centroids	Recalculate each centroid as the mean of all points assigned to it: $C_k = (1/n) * \sum(x_i)$
Step 5 — Repeat	Repeat assignment and updating until centroids stop changing (convergence).

K-Means Implementation

```

from sklearn.cluster import KMeans

model = KMeans(
    n_clusters=3,
    random_state=42
)

model.fit(X)
labels = model.predict(X)

```

Elbow Method — Finding Optimal K

The Elbow Method calculates **WCSS** (Within Cluster Sum of Squares) for different values of K. The optimal K is at the 'elbow' point where adding more clusters gives diminishing returns. Lower WCSS indicates better clustering.

Advantages vs Disadvantages

Advantages	Disadvantages
✓ Simple and fast	✗ Must specify K manually
✓ Easy to implement	✗ Sensitive to outliers
✓ Works well on large datasets	✗ Poor on non-spherical clusters
✓ Efficient for spherical clusters	✗ Sensitive to centroid initialization

4. Hierarchical Clustering

Hierarchical clustering creates a **hierarchy of clusters** and does not require specifying the number of clusters upfront. Two approaches exist:

1. Agglomerative Clustering (bottom-up)
2. Divisive Clustering (top-down)

Agglomerative Clustering

Starts with each data point as its own cluster, then repeatedly merges the two nearest clusters until all points belong to a single cluster or the desired number of clusters is reached.

Dendrogram

A **dendrogram** is a tree-like diagram that records the sequence of cluster merges. It helps determine the optimal number of clusters by cutting the tree at an appropriate height — where the longest vertical line is crossed by a horizontal cut.

Hierarchical Clustering Implementation

```
from sklearn.cluster import AgglomerativeClustering

model = AgglomerativeClustering(
    n_clusters=3
)

labels = model.fit_predict(X)
```

Advantages vs Disadvantages

Advantages	Disadvantages
✓ No need to specify K initially	✗ Computationally expensive $O(n^3)$
✓ Produces dendrogram visualisation	✗ Slow for large datasets
✓ Works well for small datasets	✗ Difficult to undo cluster merges

5. DBSCAN

Density-Based Spatial Clustering of Applications with Noise

DBSCAN is a **density-based** algorithm that groups closely packed points together while identifying outliers (sparse points) as **noise**. It can discover clusters of arbitrary shape.

Key Terms

Core Point	A point with at least min_samples neighbours within radius eps.
Border Point	A point near a core point but with fewer than min_samples neighbours.
Noise Point	An outlier that does not belong to any cluster (label = -1).

DBSCAN Parameters

- **eps** — Defines the neighbourhood radius. Points within eps of a core point are neighbours.
- **min_samples** — Minimum number of points required to form a dense region (core point).

DBSCAN Implementation

```
from sklearn.cluster import DBSCAN

model = DBSCAN(
    eps=0.5,
    min_samples=5
)

labels = model.fit_predict(X)
# Noise points get label = -1
```

Advantages vs Disadvantages

Advantages	Disadvantages
✓ Detects outliers automatically	✗ Sensitive to eps parameter
✓ Works on arbitrary-shaped clusters	✗ Poor on varying density clusters
✓ No need to specify number of clusters	✗ Difficult parameter tuning

6. Gaussian Mixture Model (GMM)

GMM is a **probabilistic** clustering algorithm that assumes data is generated from multiple Gaussian distributions. Unlike K-Means which assigns each point to exactly one cluster, GMM provides a **probability** that each point belongs to each cluster.

Gaussian Distribution Formula

```
f(x) = (1 / (sigma * sqrt(2*pi))) * exp( -(x - mu)^2 / (2 * sigma^2) )
```

Where:

mu = mean of the distribution

sigma = standard deviation

Expectation-Maximization (EM) Algorithm

GMM is fitted using the EM algorithm, which alternates between two steps:

- 1. Expectation Step (E-step)** — Estimate the probability that each data point belongs to each Gaussian component.
- 2. Maximization Step (M-step)** — Update the Gaussian parameters (mean, covariance, weight) to maximise likelihood.

GMM Implementation

```
from sklearn.mixture import GaussianMixture

model = GaussianMixture(
    n_components=3,
    random_state=42
)

model.fit(X)
labels = model.predict(X)
probs = model.predict_proba(X) # soft assignments
```

Advantages vs Disadvantages

Advantages	Disadvantages
✓ Flexible elliptical cluster shapes	✗ Computationally expensive
✓ Probability-based soft assignments	✗ Sensitive to initialisation
✓ Better than K-Means for overlapping data	✗ Can overfit small datasets

7. Algorithm Comparison

Summary of clustering algorithm characteristics:

Algorithm	Cluster Shape	Outlier Detection	Need K?	Speed
K-Means	Spherical	No	Yes	Fast
Hierarchical	Any	No	Optional	Slow
DBSCAN	Arbitrary	Yes	No	Medium
GMM	Elliptical	No	Yes	Medium

8. Evaluation Metrics

Silhouette Score

Measures how similar a point is to its own cluster compared to other clusters. Range: -1 to +1. Higher value indicates better-defined clusters. Score near +1 means well-separated; near 0 means overlapping; negative means misclassified.

Davies-Bouldin Index

Ratio of within-cluster scatter to between-cluster separation. Lower value indicates better clustering. 0 is the best score.

Inertia (WCSS)

Within Cluster Sum of Squares — sum of squared distances from each point to its centroid. Used primarily in K-Means. Lower inertia means more compact clusters.

Metric Summary

Metric	Range	Best Value	Algorithm
Silhouette Score	-1 to +1	+1	All
Davies-Bouldin	0 to inf	0	All
Inertia (WCSS)	0 to inf	0	K-Means

9. Real-World Applications

■ Customer Segmentation

Group customers based on purchasing behaviour, demographics, or browsing patterns for targeted marketing.

■ Image Segmentation

Separate objects or regions in images. Used in medical imaging, autonomous vehicles, and photo editing.

■ Fraud Detection

Identify suspicious transaction patterns that differ significantly from normal cluster behaviour.

■ Recommendation Systems

Group users with similar interests to make personalised product or content recommendations.

■ Healthcare

Disease classification, patient grouping, gene expression analysis, and medical imaging analysis.

10. Conclusion

Clustering is one of the most important techniques in unsupervised machine learning. Understanding the strengths and limitations of each algorithm helps in choosing the right approach for a given dataset and problem.

Algorithm	Best For
K-Means	Large datasets, spherical clusters, known K
Hierarchical	Small datasets, exploratory analysis, unknown K
DBSCAN	Noisy data, arbitrary-shaped clusters, unknown K
GMM	Overlapping clusters, probabilistic assignments needed

Clustering is widely used in machine learning, business analytics, healthcare, recommendation systems, and artificial intelligence — making it an essential tool for any data scientist or ML engineer.

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